

Soham Gugale

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Professional Summary

Mechanical Engineering Graduate student with hands-on experience in ETABS modelling, Finite Element Analysis (FEA), and data analytics. Coursework in structural design, mechanics and optimisation with proficiency in ETABS, ANSYS, Python and Power BI.

Education

Duke University, Pratt School of Engineering Aug 2025 – Dec 2026
Master of Engineering in Computational Mechanics and Scientific Computing
Durham, NC

Indian Institute of Technology, Bombay Aug 2021 – May 2025
Bachelor of Technology in Civil Engineering
Bombay, India

Work Experience

Data Science Intern May 2024 – Jul 2024
KultureHire, Puducherry, India

- Designed 3 interactive Power BI dashboards (finance, service, orders), driving **25% faster decision-making**.
- Analysed **5,000+ survey responses** using SQL (CTEs, joins), delivering insights on Gen-Z career choices.
- Automated a robust ETL pipeline that reduced overall reporting time by nearly 40% across multiple datasets.

Research Intern (Thesis on Decarbonisation) May 2023 – Jul 2023
IIT Bombay, India

Under Prof. Ajay Deshpande

- Conducted lifecycle carbon assessment for **6 infrastructure projects** in the construction sector.
- Developed simulation models estimating **18–22% carbon reduction** using alternative materials.

Structural Design Engineer Intern May 2022 – Jul 2022

S.R. Construction, Pune, India

- Designed and analysed a 4-storey RCC office building in ETABS, ensuring compliance with IS 456 and IS 1893 codes.
- Applied machine learning-based optimisation on slab thickness and reinforcement layout, reducing total material consumption by **12%**.
- Implemented regression models on seismic datasets, improving lateral drift prediction accuracy by **15%** compared to conventional methods.

Projects

Olympic Data Analysis — [Project](#)

- Built a dashboard with **120k+ records**, analysing trends across 120+ years of Olympic history.
- Designed heatmaps and rankings revealing medal distribution and gender participation patterns.

Stock Price Forecasting using Deep Learning — [Project](#)

- Trained Classification, Regression, RNN, and LSTM models on **1 year of Apple stock data**.
- LSTM achieved **90% accuracy**, lowering MSE to **0.0005** compared to 0.0013 (RNN).

AI-Based Image Denoising — [Project](#)

- Implemented DnCNN and autoencoders, improving PSNR from **20.53 (noisy) to 31.19 (denoised)**.
- Reduced training convergence time by **15 epochs** with optimisation techniques.

Skills

Programming: C++, Python, CSS, HTML, Javascript

ML: Regression, Statistical Modelling, Clustering, Deep Learning (Graphs, CNNs, NLP, LLMs)

Libraries: PyTorch, OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn, XGBoost, LightGBM

Tools: VSCode, Excel, Jupyter Notebook, Git, AutoCAD, ETABS, ANSYS

Data Analytics: Power BI, SQL

Certifications

- SQL Bootcamp (Udemy)
- Microsoft Excel (Udemy)
- Data Science with Python (Simplilearn)
- Autodesk Revit (Udemy)

Relevant Coursework

Finite Element Methods, Structural Mechanics, Structural Design, Mechanics of Materials, Solid Mechanics, Strength of Materials, Statics, Dynamics, Probability and Statistics, Numerical Linear Algebra, Engineering Design, Machine Learning, Data Structures and Algorithms